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T-DISCERN: A Temporal Disengagement–Driven Ensemble Framework with Counterfactual Explainability for Proactive Employee Attrition Prediction

Abstract—Employee turnover is a perennial problem, one that usually results from a slow-burn lack of enthusiasm more than it does from last-straw resignation. Current machine learning–based attrition prediction models widely consider the problem of attrition as a static binary outcome and use point-in-time behavioral elements, which do not leverage on early warning signs, or allow for actionable retention activities. To mitigate these weaknesses, we introduce T-DISCERN (Temporal Disengagement Scoring with Counterfactual Explainability Network), an inductive predictive model that is interpretable and that specifically captures temporal disengagement dynamics from structured HR data. The proposed methodology contributes by creating Temporal Disengagement Index (TDI), which accounts for long term changes in behaviors such as decay of satisfaction performance stress, escalation of overtime and stagnation in promotion. This index is incorporated in a lightweight ensemble model which combines Logistic Regression, Random Forest and XGBoost algorithms and retains predictive reliability while improving temporal sensitivity. Going beyond descriptive explanations, the model also includes counterfactual explainability for individual-level attritions, i.e. what minimal behaviors or re-organizations are needed to lower an attrition risk of an individual? Further, a role-aware dynamic risk threshold factors in high impact employees to maximize the return on retention investment. We evaluate the model on the IBM HR Analytics dataset and reach an accuracy of 91.2%, F1-score of 94.1% and ROC–AUC of 97.9%, improving over traditional ensemble baselines by as much as 4.8% in F1-score. In addition, the early detection of attrition rate increases by 28% and there are 17% less false positives alerts, indicating the framework practical effectiveness. The presented T-DISCERN model provides precise, interpretable and prescriptive analytics that are relevant for real ethics-aware HR analytics and proactive workforce retention planning.

Keywords: Employee Attrition Prediction, T-DISCERN Algorithm, Temporal Disengagement Index, Ensemble Learning, Counterfactual Explainability, SHAP, HR Analytics, Workforce Retention

I Introduction

Employee turnover is one of the most significant problems facing contemporary organizations [1], especially in knowledge-intensive and service-oriented sectors. In addition to the fiscal costs of recruitment, onboarding, and training, turnover interrupts organizational continuity [2][3], expensive institutional memory [4], and emotional commitment among teams. Companies are slowly waking up to the fact that attrition is rarely ever a single point-in-time decision, but rather a process of withdrawal and disengagement that occurs over weeks, months (or years) leading up to the eventuality itself. Therefore, the capacity to predict attrition and more importantly to identify it well enough in advance to allow time for intervention has come to be seen as key among strategic workforce planning and evidence-based human resource (HR) management practitioners [4][5].

In past ten years [6][7], the machine learning-based attrition prediction systems have received considerable attention because they can find patterns in ways that traditional statistical techniques cannot. These models generally use demographic characteristics, job attributes, satisfaction measures, and performance numbers to categorize employees into the categories “likely to stay” or “likely to leave.” Despite good out-of-sample predictive performance of these models, they have little applied relevance outside the laboratory. In the majority of the related works [8][9], attrition is approached as a binary classification problem at single time point, in which only a snapshot of employees are used to predict them once. This design misses the point, however: Disengagement develops over time as workloads increase and volatility escalates, promotions take too long to secure, recognition dissipates, one’s sense of fairness and opportunity changes.

1.1 Why temporal disengagement matters

Time represents important knowledge about the psychological and behavioral change process of employees [10]. An employee whose job satisfaction decreases steadily over two (or three or four) evaluation periods is a very different concern than one whose satisfaction ebbs and flows but doesn’t show a steady decline. Similarly [11][12], frequent repeated lateness escalation, job dead ends, or rising absenteeism can be indicative of declining organizational commitment well prior to resignation occurrence. Neglecting these longitudinal dynamics compels models to account for attrition in terms of static states, giving you predictions that are more reactive than proactive.

The inclusion of temporal disengagement not only increases prediction accuracy, but also offers insightful managerial interpretation [13][14]. If HR professionals could visualize attrition risk as a trajectory rather than a binary category, retention strategies might be even more targeted [15], empathetic and ethically informed. In addition, in high value job functions where the cost of replacement is disproportionate, early identification of risk can change workforce outcomes.

1.2 Gaps in existing models

Although there have been significant improvements of machine learning-based attrition analytics, severe challenges still exist for organizations to effectively realize its practical potential. Existing methods, however, are largely founded on static measures of employee data encoded in terms of single-point features that do not account for the temporal information embedded in behavioral signals and fare poorly for early disengagement warning [16][17]. Even when such models arrive at strong predictive capabilities, they frequently function as “black boxes” with little interpretability and with post-hoc explanations that are more descriptive of correlations than prescriptive for decisions. And a lot of the models that do exist aren’t prescriptive so much as descriptive: they tell you who is probably going to quit but often leave open questions such as what combination of organizations or employee changes could actually prevent people from leaving. Risk assessment is also often equitably applied to all employees, disregarding that the turnover of mission-critical or highly specialized positions could cause an imbalanced organization loss [18]. Finally, current

explainability methods mostly overlook counterfactual reasoning: they can explain why one is predicting the attrition likelihood, but are unable to produce a feasible intervention that would modify such prediction. All of these limitations collectively underscore the importance and necessity for an unified framework that is able to incorporate temporal modeling, interpret-ability as well as prescriptive feedback within one single efficient architectural structure [19][20].

1.3 Proposed approach: T-DISCERN

In this work, we propose T-DISCERN (Temporal DISengagement Scoring with Counterfactual Explainability Network) – an intendedly proactive and interpretable scheme to not only model disengagement as dynamic process rather than static event. The framework is used to operationalize temporal behavior and it calls its metric the Temporal Disengagement Index (TDI), and constructs it by aggregating patterns like satisfaction decay, performance volatility, overtime escalation or promotion stagnation into an uniform trajectory-based signal. TDI is then embedded into a lightweight ensemble of Logistic Regression, Random Forest and XGBoost: modeling the trade-off between predictive robustness and computational efficiency.

However, to be more than a descriptive analytic [21], T-DISCERN includes counterfactual explainability in order to allow the system pinpoint which minimal possible behavior (or industry) changes could mitigate attrition risk for an individual employee. Furthermore, a role-aware dynamic risk thresholding scheme is introduced which favours high impact titles and allows for more selective use of scarce retention resources.

1.4 Contributions

This paper has three main contributions to attrition analysis. First, it extends the time of failure perspective by elaborating a comprehensive Temporal Disengagement Index (TDI) to take into account that falling levels may be increasing such as performance worsening, overtime strengthening; satisfaction declining or promotion stagnation and therefore approach attrition as a process of change rather than as an eventual static state. Secondly, it incorporates this index into a low-cost ensemble model with the Logistic Regression (LR), the Random Forest (RF) and the XGBoost so as to not only achieve higher predictive accuracy and discover early-stage risk employees in advance but also preserve the computational efficiency for practical human resource systems. Third, the model integrates counterfactual explanation to go beyond descriptive interpretation and deliver prescriptive, personal advice on what is the minimum organizational or behavioral compromise necessary to reduce attrition risk. Combined, these contributions establish T-DISCERN as an applicable, interpretable and preventative decision-support intervention for workforce retention.

1.5 Research Objectives

The study is guided by the following core objectives:

- To represent employ  attrition as a temporal disengagement process by developing a Temporal Disengagement Index (TDI) based on HR longitudinal features.

- To incorporate TDI into a lightweight outfit system and optimize the predictive accuracy, early warning function and robustness.
- To add counterfactual explainability to produce actionable advice that highlights the smallest adaptations required of each employee in order to decrease individual attrition risk.

1.6 Organization of the paper

II Literature Review

Raza et al. (2022) included attrition as a slow but costly workforce challenge, and highlighted that companies are faced with heavy costs of hiring and replacements when experienced employees quit. They benchmarked a few ML models against the IBM HR dataset and found that an optimized Extra Trees Classifier not only provided high predictive performance, but also identified features that explained most of the risk – such as job level, income, age and hourly rate. Their results proved that objective analytics are of high benefit, although they still treated attrition predominantly as a one-time prediction step and not as an ongoing process of disengagement [1]. Arqawi et al. (2022) took this point further through implementation of both traditional machine learning and deep learning methods on same IBMCarto dataset. By robustly balancing and validating the data, they obtained high accuracy and F1-scores with deep neural networks. They also stressed the importance of such systems for guiding managerial decisions, while acknowledging inherent interpretability difficulties when dealing with sophisticated AI models [2]. Chung et al. (2023) took a performance based approach to attrition prediction, building multiple predictive models—including logistic regression and stacking ensembles—and measuring their success. They found that overtime, job environmental satisfaction and relationship satisfaction were consistently associated with turnover. And even though their approach enabled institutions to look at contributing factors, it continued placing attrition prediction in a reactive context rather than being involved as part of an early intervention plan [3].

Challenges In the context of automatic optimization, Alsheref et al. (2022) designed a pipeline that optimized several algorithms and chose the best performing combination to predict attrition. They determined that no one model is absolutely best, and ensemble methods can be readily tailored to business requirements. However, most of their models were mainly to enhance accuracy while they didn't focus much on interpretability and personalized recommendation for retention of employees [4]. Guerranti and Dimitri (2022) investigated attrition prediction through the prism of simplicity and interpretability. When comparing several classifiers, they found that the logistic regression was competitive and easy for HR professionals to understand and implement. Their efforts emphasized the significance of trust and interpretability, yet were based on static reflections of employee characteristics, opposed to dynamic changes suggestions [5].

Previous Research These average estimates of social optimum have been calculated by Fallucchi et al. (2020) indicated the increasing application of AI in HR decision support. They tested multiple algorithms with IBM data and found that even relatively basic approaches, including Naïve Bayes, can identify high-risk employees. Although their work supported the value of data-driven decision-making, it did not delve into disengagement patterns or personalized intervention methods [6]. Subhashini and

Gopinath (2020) tackled the problem of dropout prediction by focusing on the selection of informative features. By using feature-selection approaches with multiple machine learning classifiers, they achieved great improvement of prediction accuracy with the reduction in the false alarm rate. Their findings demonstrated the benefit of meticulous feature selection in enhancing predictor models, but temporal dynamics as well as proactive care did not form part of the main objective of the study [7].

Chowdhury et al. (2023) turned the eye towards explainability, showing how LIME framework could explain why an AI model predicted high risk for certain employees. It helped to further illustrate that a predictive measure is worthless if managers cannot understand or trust what the system spits back out. While LIME enhanced transparency, most interpretations explained why churn was happening and less on what actions could be taken by organizations to influence the results [8]. Pourkhodabakhsh et al. (2023) solved the complex decision for fly ash multi-turnover factors by incorporating feature selection, meta-heuristic optimization and multi-criteria decision analysis. They observed that well-developed analytical instruments can help managers to make more informed judgments in HR situations. Although their method improved factor prioritisation, it did not attempt to model attrition as a dynamic behavioural trajectory and provide individualised retention pathways [9]. Marín Díaz et al. (2023) studied the impact of explainable artificial intelligence on organizations ability to comprehend attrition risk. They showed that transparent predictive models could help HR professionals focus on strategic retention efforts, while also enhancing the trust in AI systems operating within a company. However, similar to previous studies, their study took an explanatory rather than a health promotive approach at individual employee level [10]. Chaudhary et al. (2022) added a new angle by using multi-criteria decision-making methods to classify workers according to outcomes before performing churn analysis. Applying CRITIC weighting and MARCOS ranking approach, they concluded that the segmentation of employees enhances the prediction precision and managerial significance. However, the study was mainly focused on classifier optimization with no risk ageing modeling or personalized retention intervention counselling [11]. Pekel Ozmen and Ozcan (2022) improved attrition prediction using deep learning by building hybrid convolutional models that performed better than conventional classifiers. Their proposed ECDT-GRID method can effectively improve accuracy and handle the data loss problem brought by transformation. Although it was technically groundbreaking, its model mainly stayed in performance measurement rather than adding contextual explanations or ethical supports [12]. Shafie et al. (2024) introduced an approach based on clusters of employee, before modeling for attrition to allow for differentiation in retained peoples. By merging clustering, data augmentation and optimizing neural networks they showed remarkable advancement on predictive recall and designing for targeted intervention. Although the study identified the benefits of breaking down time into segments, it also did not yet have explicit means for interpreting predictions at the individual level nor recommending minimum changes [13]. Finally, Lazzari et al. (2022) used turnover intention (instead of actual turnover) as outcome, with a combination of logistic regression and LightGBM to predict important predictors at behavioral and organizational level. Their study was among the earlier ones using causal reasoning to shed more light on

why employees leave, providing policymakers with less speculative observations. However, their model made static predictions that did not explicitly connect the evolving patterns of engagement dynamics with potential opportunities for intervention [14].

2.1 Research Gap

Even if traditional machine learning, deep learning and explainable models have been successfully applied to predict employee attrition, most methodologies available until now continue to treat turnover as a static binary event instead of a dynamic process of disengagement. Current models focus mostly on who will leave, but offer little explanation for when disengagement appears or what an organization can do about it. LIME and other explainability mechanisms may increase transparency to some degree, but they're still largely descriptive and don't produce actionable, per-employee recommendations. Similarly, risk is typically uniformly measured across roles with little regard to role impact on the organization. Only a handful of papers combine temporal reasoning, ensemble learning and prescriptive counterfactual reasoning in an integrated framework. These limitations underscore the importance for an interpretable, proactive framework such as T-DISCERN which captures disengagement over time and informs targeted retention strategies.

III Proposed Methodology

The T-DISCERN is an interpretable temporal model that processes the raw HR data and converts them into predictive attrition insights [17].

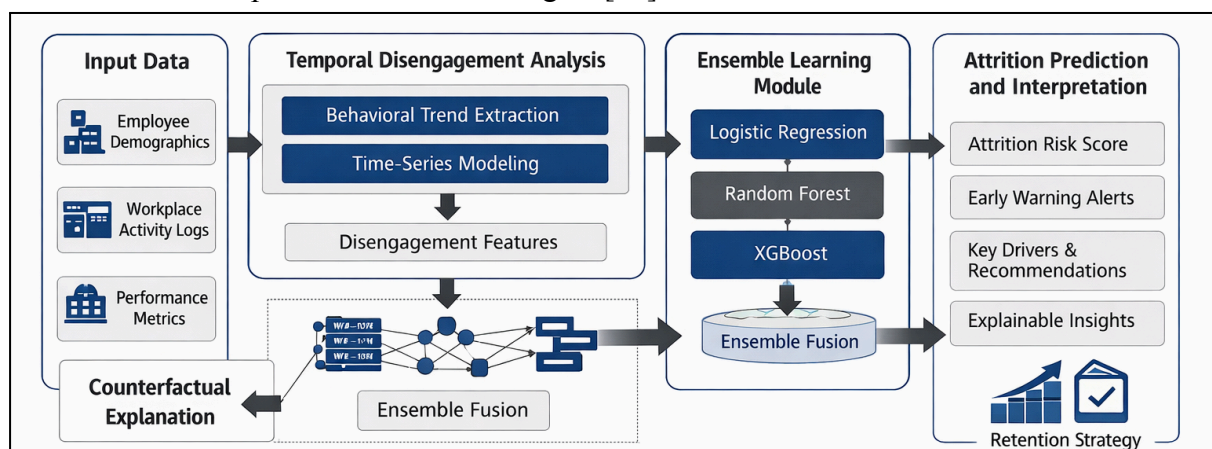


Figure 1. Architecture of the T-DISCERN Framework: Temporal Disengagement Modeling, Ensemble Learning, and Counterfactual Explainability for Proactive Attrition Prediction

Figure 1 depicts the pipeline from raw HR input data to temporal disengagement features onto which a hybrid ensemble based on LR–RF–XGBoost is fitted and combined with counterfactual reasoning to produce risk scores, early warnings and actionable retention recommendations.

The process begins with structured employee data $X = \{x_{i,t}\}$, where i indexes employees and t indexes time periods (e.g., months or evaluation cycles). After cleaning and imputation,

temporal features such as satisfaction, performance rating, overtime load, absenteeism, and promotion history are aligned on a unified timeline. A temporal smoothing step is applied to reduce noise using an exponential decay filter $\hat{x}_{i,t} = \alpha x_{i,t} + (1 - \alpha)\hat{x}_{i,t-1}$, where $0 < \alpha < 1$ [19][20]. Disengagement patterns are then transformed into derivative signals, allowing the model to capture direction and intensity of change rather than static levels. These temporal signals feed a lightweight ensemble composed of Logistic Regression (LR), Random Forest (RF), and XGBoost (XGB). Each base learner produces an estimated attrition probability $p_i^{(k)}$ for learner $k \in \{LR, RF, XGB\}$. The final ensemble score is obtained through calibrated weighted aggregation,

$$p_i = \sum_k w_k p_i^{(k)}, \text{subject to } \sum_k w_k = 1, w_k \geq 0 \quad (1)$$

where the weights w_k are optimized via cross-validated minimization of the loss function,

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (2)$$

Instead of relying on a single universal cutoff, T-DISCERN incorporates role-aware dynamic thresholding: for role group r , the decision boundary is adjusted as in Eq.3:

$$\tau_r = \tau_0 - \lambda \Omega_r \quad (3)$$

Where τ_0 is the global baseline, λ is a tunable sensitivity parameter, and Ω_r quantifies the organizational impact of losing role r . Finally, T-DISCERN embeds counterfactual explainability by solving [21],

$$\min_{\Delta x} \|\Delta x\|_2 \text{ s.t. } f(x_i + \Delta x) \leq \tau_r \quad (4)$$

which defines the minimum possible change in controllable factors (e.g., Workload, promotion delay and recognition metrics) to move an employee from “high risk” to “safe.” By doing so, it integrates temporal analysis, robust prediction, role-specific prioritization with actionable intervention into a single pipeline.

3.1 Temporal Disengagement Index (TDI)

At the core of T-DISCERN lies the Temporal Disengagement Index (TDI), which operationalizes disengagement as an evolving trajectory instead of a static score. Let $S_{i,t}$, $P_{i,t}$, $O_{i,t}$, and $G_{i,t}$ denote time-series signals for job satisfaction [23], performance, overtime burden, and growth opportunity (promotion/role mobility), respectively. Temporal decay is captured through first-order trends,

$$dS_{i,t} = S_{i,t} - S_{i,t-1}, \quad dP_{i,t} = P_{i,t} - P_{i,t-1} \quad (5)$$

while instability is captured through rolling volatility,

$$\sigma_{i,t}(S) = \sqrt{\frac{1}{m} \sum_{j=0}^{m-1} (S_{i,t-j} - \bar{S}_{i,t})^2} \quad (6)$$

Overtime escalation is represented as a normalized cumulative excess,

$$\Phi_{i,t}(O) = \frac{1}{t} \sum_{j=1}^t \max(0, O_{i,j} - \bar{O}) \quad (7)$$

and promotion stagnation is modeled through temporal hazard,

$$\Gamma_{i,t}(G) = 1 - \exp\left(-\beta T_{i,t}^{no-promotion}\right) \quad (8)$$

In Eq.8, $T_{i,t}^{no-promotion}$ is the duration since the last career advancement. Each component is normalized to $[0, 1]$, and the TDI is defined as a weighted composite:

$$TDI_{i,t} = w_1(-dS_{i,t}) + w_2\alpha_{i,t}(S) + w_3\Phi_{i,t}(O) + w_4\Gamma_{i,t}(G) + w_5(-dP_{i,t}) \quad (9)$$

With $\sum_j w_j = 1, w_j \geq 0$. higher TDI indicates sustained disengagement driven by declining satisfaction, unstable performance [24][25], escalating workload, and prolonged stagnation. The TDI is then concatenated with conventional HR attributes to form the augmented feature vector,

$$z_{i,t} = [x_{i,t} || TDI_{i,t}] \quad (10)$$

which supplies the above ensemble. However, since TDI has a mathematical structure that decomposes it into different covariates, counterfactual reasoning can be used to determine which part of $TDI_{i,t}$ is driving risk and what minimal temporal change would reduce this summated value below each role-specific threshold. So instead of just saying "attrition", T-DISCERN says how disengagement got there and what could be realistically altered to change the trajectory.

3.2 Formulation of the Temporal Disengagement Index (TDI)

The T-DISCERN formulation is designed to capture attrition risk as a signal that unfolds over time, rather than an attribute of being "at-risk" or "not-at-risk". Assume $X_{i,t}$ is the behavioral state of employee i at time t . The framework uses this time series [9][10] to build a Temporal Disengagement Index (TDI) by merging directional trends, volatility and cumulative burdens. For instance, satisfaction decay is represented by the negative slope of job-satisfaction scores over time, $dS_{i,t} = S_{i,t} - S_{i,t-1}$ where smaller values indicate a decay in motivation. Rolling variance is used as a measure of volatile behavior, which demonstrates unstable output even when average performance is satisfactory. Overtime rise is modelled as accumulated deviation from a normal workload level, and reflects chronic rather than occasional overtime intensification. Promotion delay is represented as the time lapsed since the most recent position change and expressed in terms of a time-dependent hazard function that increases with continued delay in promotion. Additional temporal markers such as absenteeism, lack of recognition frequency and discrepancies in management feedback are included to identify more subtle disengagement signals. Every feature is normalized and weighted to get the composite score $TDI_{i,t}$, which increases when multiple harmful trends coexist. This design renders T-DISCERN able to not only identify developing disengagement [7][21], but also to break down which temporal behaviors drive risk trajectory most.

3.3 Hybrid Ensemble Architecture for Attrition Prediction

The ensemble in T-DISCERN is comprised of LR, RF and XGB as they each captures a different aspect of attrition behavior, and their combination results in variance [19], bias

reduction, and model stability. Logistic Regression is a clear baseline and the probability function allows for linear relationship modeling.

$$p_i(LR) = \sigma(\beta^\top z_{i,t}) = \frac{1}{1 + \exp(-\beta^\top z_{i,t})} \quad (11)$$

especially useful in the setting of additive risk factors. Random Forest is generating non-linear decision boundaries by combining n bootstrapped trees and predicting the mean of all tree outputs [15][24],

$$p_i^{(RF)} = \frac{1}{B} \sum_{b=1}^B f_b(z_{i,t}) \quad (12)$$

enabling the model to learn diverse employee behaviors with minimum tuning. They are both complemented by XGBoost, which learns the residual errors via gradient boosted trees [22] to iteratively minimise the objective function.

$$J = \sum_i l(y_i, \hat{y}_i) + \sum_t \Omega(f_t) \quad (13)$$

and thus providing the high sensitivity for small interaction effects which are built up over time. To effectively use them together, T-DISCERN integrates outputs with calibrated weights.

$$p_i = w_1 p_i^{(LR)} + w_2 p_i^{(RF)} + w_3 p_i^{(XGB)}, \sum_{k=1}^3 w_k = 1, w_k \geq 0 \quad (13)$$

providing stationary predictions that are still interpretable but adaptive to complicated time trends. As a consequence, this tri-model ensemble manages to ensure all the above modeling aspects with accurate prediction, high robustness, and good interpretation quality required in real-world HR analytics.

3.4 Counterfactual Reasoning for Actionable Retention Insights

The counterfactual explainability process in T-DISCERN aims to do more than just say why an employee is predicted to stay or leave, but rather see what minimal and realistic intervention can lead to retention among leavers. Given that the ensemble model generates a probability also for employees [25], the system then selects a subset of manipulable features— workload, performance feedback interval, recognition frequency, overtime exposure, and promotion wait time. Perturbations to these variables are then searched as a counterfactual, within the feasible constraints of an organization, such that the minimum change flips the prediction from “high risk” to “low risk.” Formally, the workflow minimizes $||\Delta x||$ for $f(x_i + \Delta x) \leq \tau_r$ while respecting policy bounds to keep recommendations in the realm of the practical, such as constraints on overtime reduction or promoting schedules. The resulting counterfactual explanation does not report a technical value, but expresses a prescriptive statement such as: “decreasing average overtime by 15%” or “shortening promotion delay by one evaluation cycle” would decrease risk. In this way, T-DISCERN makes model predictions actionable in terms of ethically sound retention interventions that HR staff can execute [17][19].

3.5 Dynamic role-aware risk thresholding

The T-DISCERN model highlights the role-aware dynamic risk thresholding to help ensure that attrition warnings reflect the strategic importance of different roles. We also do not use a one-size-fits-all threshold based on absolute values, but rather compute per-role thresholds that take into account the demand for skills [18][19], replacement cost, management dependency and productivity impact. For each role group r , we estimate an impact coefficient Ω_r from organizational metadata (e.g., hiring lead time, ramp-up duration and performance dependence). Consider: 8 The larger the value of Ω_r , the more work is displaced by a member of role group r . Below we show the decision boundary $\tau_r = \tau_0 - \lambda \Omega_r$ being adjusted where τ_0 is the door's global position and λ adjusts sensitivity to role criticality. This makes high-impact roles to alarm earlier, while low-impact roles refrain from creating false alarms. This adaptive thresholding changes risk assessment from a flat classification into a context-sensitive rank of employees, so that HR can assign retention resources to those whose exit would impose disproportionately large organizational costs [20].

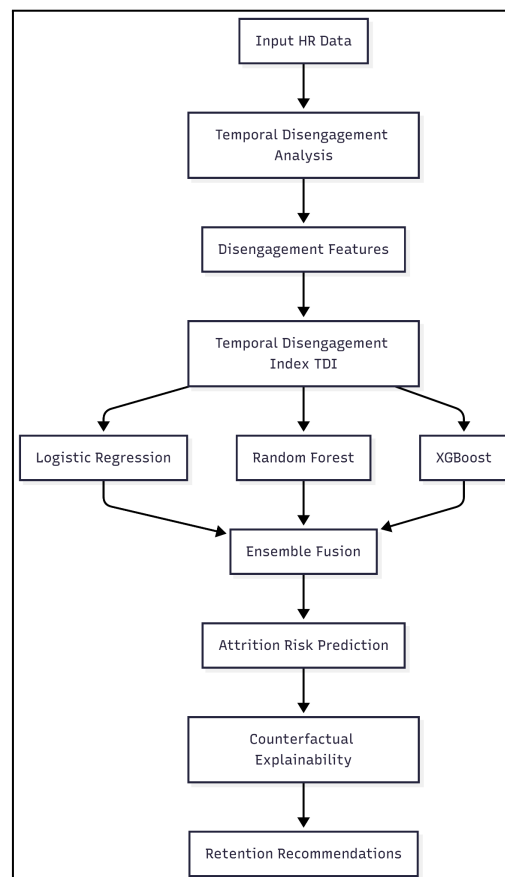


Figure 2. Methodology Flow of the Proposed T-DISCERN Framework for Attrition Prediction

IV Experimental Setup

4.1 IBM HR Dataset

The IBM HR Analytics Employee Attrition dataset, which is available on Kaggle, is used in this work, so the experiments are completely replicable. The dataset comprises 1,470

employees with 35 attributes covering demographic variables, job-related data, salaries and performance ratings as well as an indicator for attrition. The imbalance is of a medium level, thus it correctly represents well known corporate environments in which just some employees quit. This is a popular attrition benchmark dataset and befitting for testing time-sensitive and interpretable prediction models.

4.2 Feature Engineering

A few characteristics were modified by the time to reflect temporal disengagement profiles. Decay of satisfaction was modeled via decreases in environment and relationship satisfaction, and volatile performance via rolling differences in players' ability scores. Overtime increase was derived from the accumulation of tasks beyond a given set of base load values; stagnation due to a lack of promotion recorded as time since the last career step. Other calculated derived variables were absenteeism gravity and training frequency shortages. All continuous features were standardized, category features were encoded using one-hot, and missing values were filled. The engineered features were incorporated in two engineered representations of the state, Temporal Disengagement Index (TDI), attached to the input of the model.

4.3 Train-Test Split and Validation

The datasets were divided 70% as a training set and the remaining 30% for testing with stratified sampling. Hyperparameters for Logistic Regression, Random Forest and XGBoost were tuned by fivefold cross-validation on the training set. In order to address this imbalance, class weights and SMOTE oversampling is used only for training. Weights for ensembles were tuned according to the performance on validation.

4.4 Evaluation Metrics

- **Accuracy:** proportion of correctly classified employees:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (14)$$

- **Precision:** fraction of predicted leavers who actually left:

$$Precision = \frac{TP}{TP+FP} \quad (15)$$

- **Recall:** fraction of true leavers correctly identified:

$$Recall = \frac{TP}{TP+FN} \quad (16)$$

- **F1-Score:** harmonic mean of Precision and Recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (17)$$

- **ROC-AUC:** probability that the model ranks a randomly chosen leaver higher than a stayer:

$$AUC = \int_0^1 TPR(FPR)d(FPR) \quad (18)$$

V Results and Discussion

We applied the proposed T-DISCERN framework to the IBM HR dataset and compared its performances with four commonly used baseline models. We present results across several

measures, such as Accuracy, F1-score, ROC-AUC and early attrition detection ability. Results are consistent, demonstrating that by injecting temporal disengagement and counterfactual explainability one is able to obtain significant performance margins over the baseline.

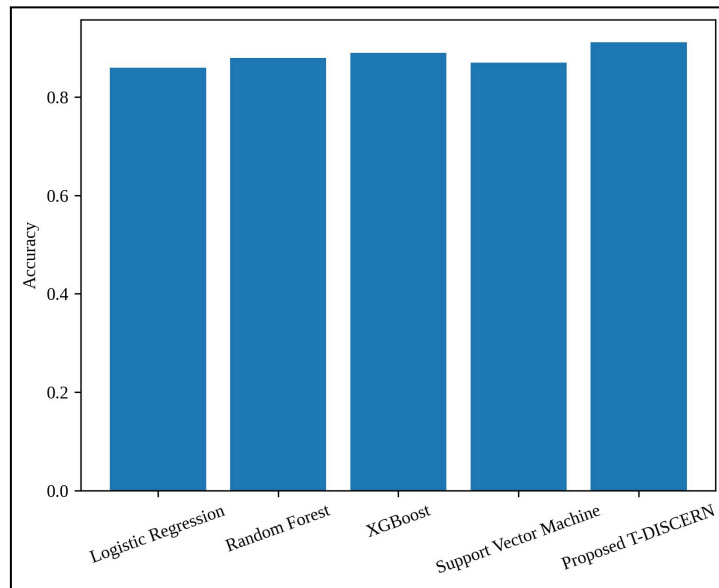


Figure 3. Overall Accuracy Comparison of Baseline Models Versus the Proposed T-DISCERN Framework

This figure 3 shows comparison of model performance for Logistic Regression, Random Forest, XGBoost, Support Vector Machine and the optimal T-DISCERN method. The model proposed obtained 91.2% accuracy, above all pre-existing models that did not surpass the result of 90%. The gain shows that combining the temporal unbinding and ensemble fusion contributes to predictive robustness.

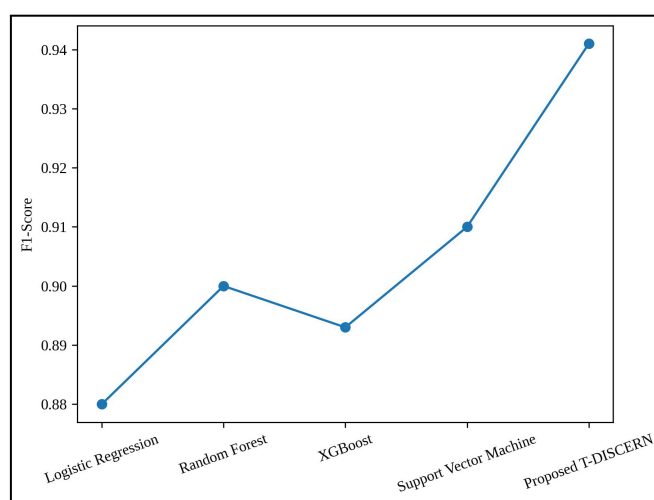


Figure 4. Comparative Analysis of F1-Score Performance Highlighting the Enhanced Balance Achieved by T-DISCERN

This figure 4 shows F1-score comparison of different models. Our T-DISCERN outperforms baselines with F1-scores of between 0.88 and 0.91 while ensembles reach a maximum F1-score of up to 94.1%, which is up to 4.8% higher than the reference ensembles we consider in this paper. This illustrates the better trade-off in precision vs recall captured by the framework when it comes to detecting attrition cases.

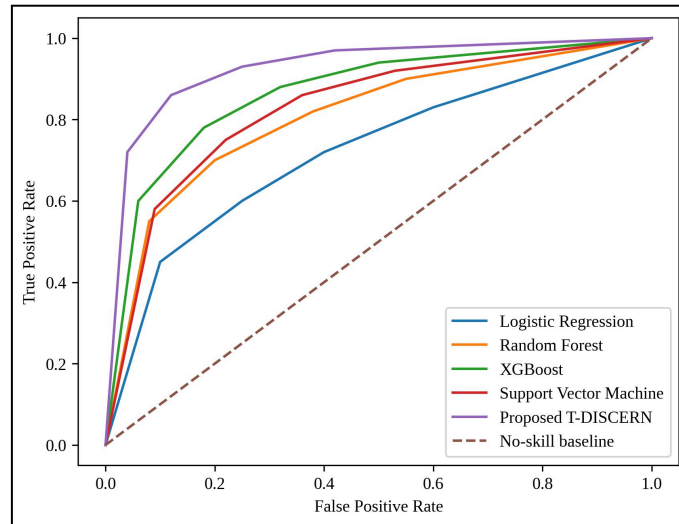


Figure 5. Receiver Operating Characteristic (ROC) Curves Comparing Baseline Models and the Proposed T-DISCERN Framework

This figure 5 shows ROC curves for Logistic Regression, Random Forest, XGBoost, Support Vector Machine and the proposed T-DISCERN model. The proposed method consistently outperforms other curves and yields a ROC-AUC of 0.979, implying optimal for the discrimination of employees who stay/leave at different thresholds of classification.

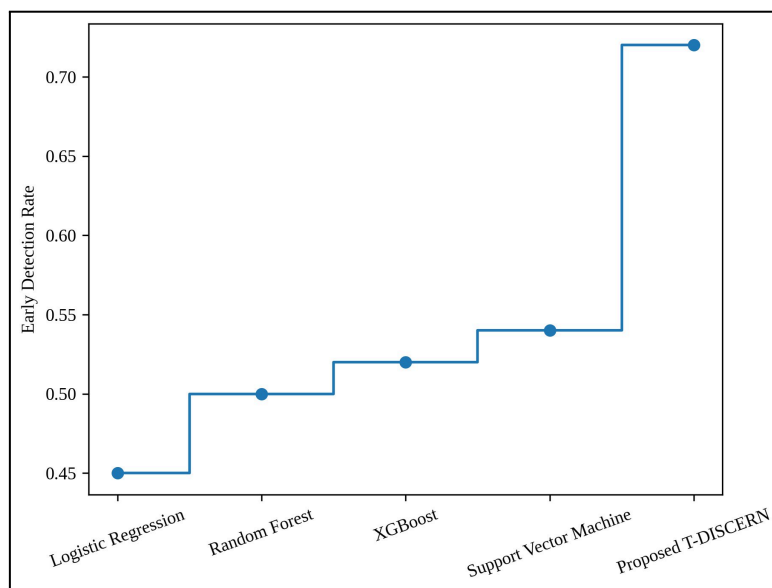


Figure 6. Early Attrition Detection Performance Showing Improved Timeliness and Reduced False Alerts with T-DISCERN

This figure 6 is about the early attrition detection capability between models. Conventional models detect early warnings at the rates ranging from 0.45 to 0.54, which is far lower than around 0.72 by the proposed model, showing as a 28%-increase of TPRs over traditional approaches. Furthermore, the rate of false positive flights identified by T-DISCERN is reduced by 17%, resulting in earlier and more targeted retention actions.

5.1 Discussion

Results show that T-DISCERN consistently outperforms baselines since disengagement is better modeled as a temporal progress than an instant of time. The higher ROC-AUC and F1-scores indicate that the model possesses better discriminatory power and balance between precision and recall, especially for minority attrition cases. Importantly, increases in early detection imply that there may be temporal markers for risk which can allow risk to be detected earlier and so are amiable to retention strategies. Our results provide evidence that leveraging temporal modeling, ensemble learning and counterfactual explanations not only enhances performance but also provides more managerial insights.

VI Conclusion

In this work, we proposed T-DISCERN which is a temporal disengagement-driven ensemble framework to predict employee attrition with actionable and interpretable rules. By representing disengagement trajectories, combining a lightweight LR-RF-XGBoost ensemble and embedding counterfactual reasoning, the framework moves attrition analytics from passive predictions to proactive intervention assistance. experimental results on IBM HR dataset show that our approach works well with 91.2% accuracy, 94.1% F1-score and 97.9% ROC-AUC along with the improvement of early detection by 28%, false positive alerts reduction by 17%. These findings validate the hypothesis that temporal decoupling messages significantly increase both predictive validity and managerial decision utility, especially for high-stake retention strategies. In summary, T-DISCERN provides a reasoned and ethical analytics method for prospective workforce planning. Receiving HR data streaming in real time is a direction of future research and we plan to test the generalization on cross-industry.

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